

Validator-Guided Repair on a Shared Initial LLM Candidate: A Preregistered Paired Evaluation for Network Configuration Safety

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Abstract—We preregistered and executed a paired confirmatory experiment (PREREG-CAND-001-VER-GVR-2026-09-01) to measure whether permitting a deterministic network-configuration validator plus one automated repair changes the rate of validator-valid executable configurations relative to leaving the identical sampled LLM candidate unguarded (`shared_initial_candidate` semantics). Using the declared generator `gpt-5-mini` (hosted adapter `openai_responses`) and deterministic `netops_validator_v5` on $N = 200$ paired intents (`completed_episode_count = 400`; `model_calls_used = 200`), every shared initial candidate passed validator checks and no repairs were attempted or applied (`repair_attempt_count = 0`; `repair_applied_count = 0`). Observed outcomes: `baseline_success = 200/200`, `guarded_success = 200/200`; paired contingency $n_{11} = 200$, $n_{10} = 0$, $n_{01} = 0$, $n_{00} = 0$; exact McNemar two-sided $p = 1.0$; paired bootstrap percentile 95% CI (seed = 1729; 10,000 resamples) = $[0.0, 0.0]$. We present artifact-grounded methodology, execution accounting, representative validator diagnostics, compact contingency and repair summaries, and a focused discussion clarifying the causal limits of the executed estimand under `shared_initial_candidate` semantics and preregistered follow-on designs required to measure repair efficacy under independent sampling, higher difficulty, or adversarial inputs.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration sequences in network operations (NetOps). Erroneous sequences, syntactic mistakes, or fabricated fields can cause service disruption or policy violations when applied to devices. A common architectural mitigation is a generate→validate→repair (GVR) pipeline: generate candidate configuration(s), deterministically validate them against intent/topology/workflow constraints, and trigger automated repair when the validator finds faults. Prior work documents verifier-guided improvements in infrastructure-as-code and related code-generation domains and recommends grounding strategies to reduce hallucination in operational tasks [1–3]. However, whether verifier-guided pipelines reduce execution-level operational risk for network configurations remains an open empirical question because prior demonstrations often measure textual correctness or IaC test suites rather than executed device-state safety in packet-level or emulator contexts [3,4].

This paper reports a fully preregistered paired confirmatory experiment that isolates the downstream causal contribution of a deterministic validator and any permitted repair acting on the

exact same sampled LLM candidate (`shared_initial_candidate` semantics). Under these semantics baseline and guarded episodes for each intent begin from the identical initial generation; any paired difference can therefore only arise from deterministic validation and a permitted repair of that same candidate. Our primary research question is: under `shared_initial_candidate` semantics, does permitting a deterministic validator plus at most one automated repair change the proportion of validator-valid executable configurations relative to leaving the same initial candidate unguarded? The contribution is an artifact-grounded report of the executed estimand with complete execution accounting, exact inferential statistics, representative validator diagnostics, and precise recommendations for preregistered follow-on designs that can measure repair efficacy when independent sampling, elevated difficulty, or adversarial inputs are employed.

II. RELATED WORK

The literature motivating verifier-guided pipelines for operational automation spans documented LLM unreliability, grounding/RAG methods, verifier-guided code/IaC repair systems, and NetOps capability studies. Surveys and empirical analyses consistently identify hallucination and factual unreliability as core risks when applying LLMs to operational tasks; these works motivate grounding and verification layers to limit unsupported or fabricated outputs in high-stakes contexts [1]. Evidence-grounded and retrieval-augmented generation (RAG) designs have been shown to reduce hallucination for operations-style question answering and document retrieval tasks in applied evaluations, improving textual factuality and traceability in AIOps and DevOps scenarios; however, these studies typically measure document/answer correctness rather than network-state executability after applying generated device commands [2].

Generate→validate→repair architectures are an active research direction in adjacent domains. Recent work on infrastructure-as-code (IaC) demonstrates that integrating verifier feedback into generation or fine-tuning loops can reduce failing IaC deployments and improve policy compliance in that domain; these method papers provide a conceptual and empirical foundation for applying verifier-guided repair to other configuration tasks but do not by themselves establish transfer to device-level NetOps execution [3]. Related program-repair and verifier-guided self-correction studies

further show that a verification step plus automated repair can raise textual/program correctness metrics, again typically measured against test suites or formal specifications rather than emulator/execution outcomes. We therefore treat these results as methodological precedent but not as direct evidence that GVR pipelines will prevent unsafe executable network states when generator outputs are enacted on device models or emulators.

NetOps-specific evaluations have studied intent translation and LLM capability for configuration generation, showing promise for mapping operator intent to device commands but emphasizing accuracy metrics and held-out textual correctness rather than standardized execution-level safety or emulator-based validation [4]. Broad AIOps surveys and reviews highlight a persistent gap: many capability benchmarks and capability-oriented evaluations do not include standardized execution-aware safety metrics, and they call for improved validator fidelity and execution-level benchmarks to bridge generation to operational safety [5].

Putting these lines of work together suggests two cautions that guided our design. First, verifier-guided improvements observed in IaC and program-repair domains motivate testing the same architecture in NetOps but do not substitute for execution-level experiments: transferability is an empirical question. Second, experiments that conflate generator sampling variability with downstream validator/repair effects make causal attribution difficult; a clearly specified estimand and sampling semantics are necessary to isolate whether validation/repair changed outcomes for the same generated candidate versus whether improved first-pass sampling reduced errors. These considerations led us to the preregistered `shared_initial_candidate` paired design used in this study, which isolates downstream validator/repair effects on identical generator outputs while acknowledging that independent-sampling designs are needed to evaluate generator-level error reduction.

III. METHODOLOGY

Preregistration and primary estimand. We preregistered the study as PREREG-CAND-001-VER-GVR-2026-09-01 and froze the primary estimand and analysis before any model calls. The primary estimand is the `paired_success_rate_difference_guarded_minus_baseline`: the per-intent paired difference in the proportion of validator-valid executable configurations under `shared_initial_candidate` semantics. The preregistration explicitly set `generation_semantics = "shared_initial_candidate"` (exactly one sampled initial generation per pair), `task_count = 200`, `planned_episode_count = 400`, and `maximum_repair_calls_per_task = 1`. The primary statistical test specified was the exact two-sided McNemar test, accompanied by a paired bootstrap percentile 95% confidence interval (`bootstrap_seed = 1729`; `bootstrap_resamples = 10000`) for the paired difference.

Task bank, deterministic generation, and contamination controls. A deterministic task generator produced a man-

ifest of 200 synthetic `intent_configuration_repair_v1` tasks. Each manifest entry records `initial_state`, `required_sequence`, difficulty annotations (e.g., "very_hard", "extreme"), and a `generation_seed` to permit deterministic replay. Preexecution contamination checks were performed per the preregistration: SHA-256 exact-match detection across repair and corpus artifacts and n-gram similarity screening. No exact-match contamination exclusions were raised for the confirmatory sample; contamination artifacts and the task manifest are archived in the evidence bundle for independent verification. The preregistration defined explicit exclusion rules (exact-match exclusions moved to an exploratory leaked bucket) and a missingness policy (`complete_pair_primary`) for handling simulator or execution failures.

Model calls, sampling semantics, and provider audit. Execution used the declared generator labeled `gpt-5-mini` via the hosted adapter `openai_responses`. The preregistered execution contract required one shared initial generation per pair; the execution manifest records `model_calls_used = 200`, `hosted_model_call_event_count = 200`, `completed_hosted_model_call_event_count = 200`, and `cache_miss_count = 200`. The archived `model_configuration.json` declares `requested_model = "gpt-5-mini"` and `temperature = null/unset`; the preregistration and manuscript explicitly note that `null/unset` is not evidence of deterministic model sampling and does not imply `temperature = 0`. Provider audit metadata (`stage_counts` showing `shared_initial_generation = 200` and no cross-condition cache reuse) were checked as part of deterministic reconciliation to confirm the intended `shared_initial_candidate` semantics were followed.

Deterministic validator, scoring rules, and repair policy. The study used `deterministic_netops_validator_v5` with `scoring_rule = "full_intent_constraint_validation."` The validator ingests the generated candidate commands, parses them into structured per-interface field updates, applies the task's `workflow_policies` and `required_sequence` constraints, and produces a structured validator trace. Each per-episode scoring artifact includes fields such as `normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_codes`, `violation_count`, `validator_id/version`, `validation_before` and `validation_after` subtables, a `repair_applied` boolean flag, and a score with a `score_reason_code` (for example, "VALID_CONFIGURATION"). In guarded episodes the policy permitted at most one automated repair attempt (`maximum_repair_calls_per_task = 1`); if invoked, the repair would produce a revised candidate that the validator would revalidate and record in the trace. Under `shared_initial_candidate` semantics the same initial sample was provided to both baseline and guarded episodes; only the guarded arm could alter that identical candidate via the permitted repair.

Outcome mapping, missingness, and multiplicity. `Validator.valid` (`validator.valid == true` in the `validation_after` subtable) was mapped to a binary "executable/valid" outcome for

each episode. The preregistration defined the primary missingness policy `complete_pair_primary` (exclude incomplete pairs from the confirmatory McNemar analysis) and a sensitivity plan that treats any missing or failed episode as unsafe (failure-as-zero). Multiplicity control: a single confirmatory McNemar test was preregistered (no multiplicity correction for the primary test); exploratory subgroup analyses (e.g., by declared difficulty pattern) were labeled exploratory and, if reported, were to be corrected using Benjamini–Hochberg FDR at 0.05.

Deterministic reconciliation and audit checks. The methodology required deterministic reconciliation of episode accounting after execution. Reconciliation checks included verifying `planned_episode_count = 400`, `completed_episode_count` equality, symmetry of `shared_initial_generation` counts, `provider_call_audit` totals (`hosted_model_call_event_count` and cache statistics), and consistent pairing (`complete_pair_count = task_count`). All deterministic consistency checks (provider call audit, episode accounting, and pair accounting) were preregistered as required for the primary analysis to proceed. The analysis executor was instructed to read raw scoring artifacts and compute the paired contingency table exactly from per-episode `validator.valid` outcomes; bootstrap resampling used the archived seed and resample count for reproducibility.

Analysis pipeline. The analysis plan specified `paired_binary_analysis_v1` to compute the contingency table (n_{11} , n_{10} , n_{01} , n_{00}), run the exact two-sided McNemar test, and compute a paired bootstrap percentile 95% CI (`bootstrap_seed = 1729`; `resamples = 10000`) for the paired difference. The analysis artifacts generated (`analysis/condition_summary.csv`, `analysis/paired_contingency_table.csv`, `analysis/missingness_summary.csv`) were archived and hashed. Any deviations from the preregistration (for example, contamination exclusions or incomplete pairs) would be documented and reported; no such deviations occurred in this run. The methodology explicitly avoided human adjudication of validator outcomes: per preregistration, all scoring was deterministic and machine-recorded, and the primary analysis used those machine labels without human relabeling.

IV. RESULTS

Execution accounting and deterministic reconciliation. Execution completed per the manifest: `planned_episode_count = 400`, `completed_episode_count = 400`, `failed_episode_count = 0`, `model_calls_used = 200` (execution/manifests archived). Analysis `missingness_summary` reports `complete_pairs = 200` and zero failed or unscorable episodes. Deterministic reconciliation shows `stage_counts = {shared_initial_generation: 200}` and `cross_condition_cache_key_reuse_observed = false`; all deterministic consistency checks passed.

Primary outcomes and contingency. Condition summaries: `baseline_successes = 200/200` and `guarded_successes = 200/200` (`success_rate = 1.0` each). Paired contingency: $n_{11} = 200$, $n_{10} = 0$, $n_{01} = 0$, $n_{00} = 0$. Exact McNemar two-sided

$p = 1.0$. Paired bootstrap (seed = 1729; 10,000 resamples) percentile 95% CI for the paired difference = [0.0, 0.0]. These artifacts are archived as `analysis/condition_summary.csv` and `analysis/paired_contingency_table.csv`.

Repair activity and validator diagnostics. `Repair_attempt_count = 0` and `repair_applied_count = 0` across the confirmatory sample: the validator flagged no violations on any shared initial candidate and therefore the permitted guarded repair path was never invoked. Representative per-episode JSON artifacts (examples archived under `execution/scoring/`, e.g., `task-000004-baseline.json` and `task-000004-guarded.json`) show identical `shared_initial_candidate` content and identical `validation_after.normalized_configuration` values. Example shared initial candidate (task-000004):

```
interface edge2 admin down interface edge2 vlan 40 interface edge2 admin up interface edge1 admin down
```

For that example `parsed_command_count = 4`, `required_command_count = 4`, and `validator.valid = true`. Similar diagnostics hold for sampled representative tasks (task-000005, task-000006).

Contamination and provider audit summary. Preexecution contamination checks flagged no exact matches; `analysis/contamination_summary.csv` is empty. Provider audit recorded `hosted_model_call_event_count = 200` and `cache_miss_count = 200`. No infrastructure failures or retrials occurred; `analysis/analysis_log.jsonl` documents the bootstrap settings used (`bootstrap_resamples = 10000`; `bootstrap_seed = 1729`) and that paired analysis completed successfully.

Compact repair summary (explicit). Table 2 (below) provides the preregistered reviewer request summary derived from archived artifacts: `total_pairs = 200`; `repair_attempt_count = 0`; `repair_applied_count = 0`. The absence of any validator detections fully explains the observed degenerate contingency: with no validator detections, the guarded repair path had no opportunity to alter outcomes.

V. DISCUSSION

Interpreting the executed estimand. The `shared_initial_candidate` semantics were intentionally chosen to isolate the causal effect of deterministic validation and any permitted repair acting on the exact same generator output. The executed estimand therefore answers a narrowly defined causal question: when baseline and guarded episodes begin from the identical sampled candidate, can deterministic validation and at most one automated repair change the `validator-valid/executable` outcome? The observed null effect (paired difference = 0.0) shows that, for this task bank and this single sampled candidate per pair, the validator found no violations and the permitted repair path was never exercised. Importantly, this does not evaluate whether guarded pipelines reduce first-pass generator error rates under independent sampling or different model temperatures; such claims would require a different preregistered estimand and execution.

Threats to validity (artifact-grounded). Internal validity: the paired `shared_initial_candidate` design eliminates sam-

pling variability across arms and thus preserves attribution of any observed repair effect to validator/repair alone; however, this design precludes measurement of generator-level effects. Construct validity: `deterministic_netops_validator_v5` enforces syntactic and workflow-policy checks and maps results to a boolean valid flag; it is an abstraction that does not execute vendor CLI on physical hardware nor perform packet-level emulation—dynamic protocol/timing effects are therefore not represented. External validity: results are conditioned on one declared generator (`gpt-5-mini` via adapter `openai_responses`), a synthetic deterministic task bank, and one validator implementation; generalization beyond these artifacts is unsupported. Conclusion validity: zero discordant outcomes preclude inference about repair benefit magnitude under different sampling regimes because the guarded path was not invoked.

Operational implications and preregistered follow-on designs. The null result here should not be read as evidence that GVR pipelines are unnecessary in practice. Instead it demonstrates an estimand boundary: under `shared_initial_candidate` semantics and the executed task bank the guarded pipeline had no opportunity to change outcomes. To empirically evaluate repair efficacy in operationally relevant regimes, we preregistered and recommend the following estimand/design modifications (each reuses archived artifacts where applicable): (A) remove `shared_initial_candidate` semantics so baseline and guarded arms receive independent first-pass samples (this permits the repair path to act on generator faults but requires a revised paired/unpaired analysis plan to separate generation variability from repair effects); (B) select or synthesize tasks labeled "extreme" difficulty or inject adversarial perturbations to raise expected validator failure rates and thereby exercise repair logic; (C) preregister non-null temperature values and temperature sweeps to evaluate sample diversity versus validator-failure tradeoffs; (D) increase validator fidelity by integrating vendor CLI semantics or packet-level emulation so repairs are evaluated against dynamic runtime behavior; (E) expand to multi-model sampling to assess generality across generator families.

Reproducibility and artifact utility. All per-episode scoring JSON artifacts include validator traces with `validation_before/validation_after` subtables, `normalized_configuration`, `parsed_command_count`, `required_command_count`, and `repair_applied` flags permitting independent verification that repairs did not run. Execution and analysis artifacts (`execution/raw_results.jsonl`, `execution/task_manifest.jsonl`, `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`) are archived in the evidence bundle and enable replay of the exact `shared_initial_candidate` sampling. The `model_configuration.json` records `declared_model_version = "gpt-5-mini"` and `temperature = null`; `null/unset` must not be interpreted as deterministic sampling. The master prompt immutable reference is archived (see Disclosure). These artifacts are sufficient to reproduce the executed estimand and deterministic reconciliation.

VI. LIMITATIONS

- `Shared_initial_candidate` semantics: the design produced exactly one sampled initial generation per intent and reused it across baseline and guarded conditions; this measures validator/repair effects only and cannot detect differences caused by independent per-condition sampling or prompt engineering.
- Temperature unset (`temperature = null`) in the archived model configuration: `null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`; sampling nondeterminism may still have occurred during the single sampled generation.
- Single model and single hosted provider: results reflect `gpt-5-mini` under the archived adapter; generalization to other models or model families is unsupported by these artifacts.
- Synthetic task bank and validator abstraction: tasks were synthetic `'intent_configuration_repair_v1'` items and the deterministic validator is an abstraction; realistic vendor CLI idiosyncrasies, emergent runtime interactions, and hardware effects were not executed on live devices.
- No observed validator failures: all initial candidates were validator-valid, so the experiment could not assess the repair step's effectiveness; the null effect therefore reflects the task bank and sampling semantics rather than evidence that GVR is unnecessary in general.

VII. CONCLUSION

We executed a fully preregistered paired confirmatory experiment that measures the downstream effect of a deterministic validator plus an allowed single automated repair on the exact same sampled LLM candidate per intent (`shared_initial_candidate` semantics). Using the declared generator `gpt-5-mini` and `deterministic_netops_validator_v5` on 200 paired intents, every shared initial candidate passed validation and no repairs were attempted or applied; guarded and baseline outcomes were identical for the executed estimand (paired difference = 0.0; McNemar $p = 1.0$; paired bootstrap CI = [0.0, 0.0]). This artifact-grounded null result demonstrates an estimand boundary: under `shared_initial_candidate` semantics and the executed task bank the guarded pipeline could not be exercised and therefore could not change outcomes. It does not imply that GVR pipelines are unnecessary generally. Preregistered follow-on experiments that (1) remove `shared_initial_candidate` semantics, (2) increase task difficulty or inject adversarial inputs, (3) set explicit sampling parameters, and (4) raise validator fidelity are required to empirically evaluate repair efficacy under realistic sampling variability and adversarial conditions. The archived artifacts and reconciliation files enable exact reproduction of the executed estimand and provide a secure base for precise preregistered extensions.

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One-line reiteration of the primary estimand limit: the preregistered primary estimand intentionally used `shared_initial_candidate` semantics; evaluating per-condition independent sampling requires a different preregistration and analysis plan (explicitly recommended above).

DISCLOSURE STATEMENT

Disclosure (concise): The human Research Director selected the broad topic family (Generative AI / LLMs for NetOps) before the autonomy lock. After the autonomy lock the autonomous system performed literature review, experiment selection, preregistration (PREREG-CAND-001-VER-GVR-2026-09-01), code generation, execution, deterministic validation, automated analysis, and manuscript drafting without manual human editing of technical content. Declared model used in execution: `gpt-5-mini` via hosted adapter `'openai_responses'` (execution used `model_calls_used = 200`; archived `model_configuration.json` records `temperature = null/unset`; `null/unset` is not evidence of deterministic sampling and does not imply `temperature = 0`). Generation semantics executed: `shared_initial_candidate` — exactly one sampled initial generation per intent was produced and deterministically reused for both baseline and guarded conditions. Validator: `deterministic_netops_validator_v5` scored candidates using `'full_intent_constraint_validation'` and a single automated repair iteration was permitted in guarded episodes; in this run the validator flagged no violations and no repairs were applied (`repair_attempt_count = 0`; `repair_applied_count = 0`). Execution accounting: `completed_episode_count = 400` (200 paired intents) completed without preregistration deviations. Master prompt immutable reference: `provenance/master_prompt.txt` — SHA-256: `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acd1582bb39b15ba`. Pre-lock infrastructure development and rehearsal occurred but pre-lock artifacts were not used as empirical evidence in this final run. After the autonomy lock no human manually rewrote technical results, manuscript text, figures, or references.

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